

A PUF and Fuzzy Extractor-Based UAV-Ground Station and UAV-UAV Authentication Mechanism With Intelligent Adaptation of Secure Sessions

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Abstract—It is crucial that communication between an unmanned aerial vehicle (UAV) and the ground station (GS) be secure, and both devices should mutually authenticate each other to ensure that an adversary cannot obtain communicated information. Moreover, dynamically adapting the session time of an authenticated session can decrease the idle time of a session and consequently reduce the window of opportunity for an adversary to interfere with the communication link. In light of these considerations, we design a *physically unclonable function (PUF)* and *fuzzy extractor*-based UAV-GS authentication mechanism called *UAV Authentication with Adaptive Session (UAAS)*. In UAAS, both UAV-GS and UAV-UAV authentication are two-way. We use a *Thompson Sampling (TS)*-based approach to intelligently adapt the duration of a session. Both formal and informal security proofs are presented along with a computation and communication cost analysis to analyze the performance of UAAS. It is noted that UAAS is secure against various well-known attacks and has a lower communication cost than several baseline mechanisms. Due to the noise reduction that occurs in PUFs, the computational cost of UAAS is higher than that of baselines that ignore noise. Also, our simulation shows that UAAS significantly outperforms baselines when it comes to network performance.

Index Terms—UAV, mutual authentication, physically unclonable functions, fuzzy extractor, session time adaptation.

I. INTRODUCTION

RECENTLY, unmanned aerial vehicles (UAVs) have become essential for remote access since they are small in size to fly over the regions where humans cannot possibly reach, and thus UAVs are a way to monitor unreachable locations. UAVs, also known as ‘drones’, are being used for providing services in both military and civilian applications, such as environmental and infrastructure monitoring, surveillance, vehicular traffic management, inventory tracking, and disaster control and

response [1]. The network consisting of UAVs is an unmanned aerial system (UAS), which is composed of one or multiple UAVs and a ground station (GS) [2]. Due to their advantages and wide range of applications, UAVs have been integrated with the Internet of Things (IoT), termed as the Internet of Drones (IoD), to help increase capacity, dynamicity, and scalability of the IoT paradigm [3], [4].

Since the wireless medium is insecure and can suffer attacks, such as masquerade attacks, man-in-the-middle attacks, denial-of-service (DoS), replay attacks, and node capturing, security is always a critical issue in UAV communications [5]. Due to both strategic and financial information involved in UAV applications, UAVs are vulnerable to the aforementioned attacks that target the wireless link and cyber and/or physical systems [6]. For example, in a military system, an adversary UAV may try to authenticate itself to the GS as a legitimate UAV. As a result, the adversary can obtain secret information about both the GS and legitimate UAVs and send spiteful information to the GS. Therefore, to guarantee the cybersecurity of the IoD, designing security mechanisms for UAVs is of paramount importance for UAV-assisted networks [7].

Toward UAV security, it is required to achieve node authentication as the first security requirement to be fulfilled before establishing a secure session for the communication between a UAV and the GS or between two UAVs, which are termed as UAV-GS and UAV-UAV authentication, respectively. After successful authentication of legitimate UAVs, a secure session needs to be created for the communication between a UAV and the GS or between two UAVs. In addition, the session related crucial information, such as secret keys, also should be kept secure such that an adversary cannot obtain it. Due to the open air deployment, UAVs are vulnerable to node capture and cloning attacks; therefore, UAV authentication mechanisms should be able to protect secret keys from unauthorized access.

A. Related Works and Motivation

Using the physically unclonable function (PUF)-based authentication, the work in [8] proposes a mutual authentication mechanism for 5G-aided UAV networks. A UAV is authenticated at the base station (BS) based on a single challenge-response pair that is stored in the BS. In [9], a PUF-based UAV authentication model is implemented on the programmable switch with

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the objective to reduce the authentication latency. The authors in [10] propose a mutual authentication protocol for UAV swarm networks. The protocol is based on PUFs and uses a spanning tree to dynamically adapt to mobile environments. Using PUFs, the work in [11] designs a mutual authentication scheme between a BS and UAVs in UAV swarms. Although the K-Means clustering approach is utilized to form UAV clusters based on the location of the UAVs, the scheme does not focus on UAV-UAV authentication and the authors did not analyze the security against several mainstream attacks. For UAV-GS authentication, Alladi et al. [12] propose a mutual authentication protocol based on PUFs, which is further extended to UAV-UAV authentication. However, [12] does not apply fuzzy extractors to overcome the noise in PUFs and the actual challenge-response pair is stored in the GS's database, and therefore can be revealed to an adversary. The work in [13] achieves secure communications between the GS and UAVs by designing a mutual authentication protocol in which the PUF is engaged to generate the challenge-response pair to be used by UAVs and the GS.

The work in [21] designs a blockchain-assisted UAV authentication mechanism for industrial UAVs. In order to facilitate the cooperation of UAVs from different domains, Feng et al. [17] propose a blockchain-based authentication mechanism for 5G-aided IoD. Addressing the security between a drone and the GS, the work in [26] proposes an authentication scheme for securing IoD, which is based on a hash message authentication with code/secure hash algorithm. However, in [26], authentication is performed between two drones using a pre-determined time threshold before the communication with the GS. Considering the security risks in IoD networks, a security authentication protocol, which does not address inter-drone authentication, is designed in [24]. In this work, the Chinese residual theorem is applied for identity generation and verification. The authors in [25] design an authentication and secure key agreement mechanism between an accessed drone and a user with the help of a server device. Although a fuzzy extractor is used to verify the user's biometric at the login phase, PUFs are not applied and inter-drone authentication is not addressed. Using the hash functions and symmetric key primitives, the work in [27] considers flying clusters of drones and designs a user authentication scheme for IoD.

Based on UAVs' real time behavioral data collected, the work in [28] designs a model for UAV behavior prediction and identity authentication. However, [28] does not consider UAV-GS or UAV-UAV authentication, and the resistance to physical and cloning attacks needs to be analyzed. In [29], UAVs are used to improve the reliability and safety of intelligent transportation systems, where an authentication scheme combining the digital signature, hyperelliptic curve cryptography techniques, and hash function, is proposed. Considering multiple-input-multiple-output (MIMO)-based cellular connected UAVs, the physical layer security and UAV authentication are addressed in [30]. However, in [30], UAV-GS and UAV-UAV mutual authentications are not discussed. Based on a symmetric-key encryption algorithm, the authors in [31] design a secure communication mechanism to resist message interpretation, tracking, replay attack, and spoofing in communication between UAVs.

Motivation: Node authentication is the most critical security requirement when it comes to achieving UAV security, with authentication needing to be performed at both the GS and UAV ends. Designing an efficient mechanism for both UAV-GS and UAV-UAV mutual authentications is an open research challenge. In this context, PUFs can help UAVs resist physical attacks because they serve as unique identifiers for semiconductor devices. However, the PUF-based approach needs to be designed carefully, since the noise factor in PUFs can result in significant deviation in the output for the same input at different time instances. After authentication between two parties is successful, the authentication session remains valid (or active) for a period of time, known as the "session time" of authentication. It is important to dynamically set the session time once a session has been authenticated to ensure that an active session does not remain idle for a long time and thus provide an opportunity for an adversary to interfere with the communication link. Therefore, "adaptive session" denotes the dynamic adjustment of the session time of an authenticated session. Although [12] uses PUFs for both UAV-GS and UAV-UAV authentication, it does not consider the noise associated with PUFs or the intelligent adaptation of the session time while establishing an authenticated session between two parties. Dynamically adapting the session time can decrease the idle time of an active session and consequently reduce the window of opportunity for an adversary to interfere with the communication link. We present in Table I a comparison of recent related works based on their contributions and different properties, such as UAV-GS and UAV-UAV authentication, PUF use, fuzzy extractor use, mutual authentication, and intelligent adaptation of session time. The non-inclusion of these properties reveals the limitations of the related works. We also highlight in Table I that our approach considers all the aforementioned properties. To the best of our knowledge, none of the existing works consider UAV-GS and UAV-UAV authentication with dynamic session time adaptation.

B. Our Approach and Contributions

In this paper, we propose a mechanism, *UAV Authentication with Adaptive Session (UAAS)*, for UAV-GS and UAV-UAV authentication, in which a secure session is established after the successful authentication of the UAVs and GS. In the UAV-GS authentication, a *two-way* authentication, which ensures mutual authentication between a UAV and the GS, needs to be completed. Afterwards, a secure session is established between the UAV and GS, and a session key is generated for further communication between the pair. The phrase "two-way authentication" refers to the authentication process during which both the UAV and GS are authenticated by each other. This means that the UAV is verified at the GS and the GS is also authenticated at the UAV, resulting in mutual authentication. Additionally, since the information required for authentication is exchanged between the UAV and the GS, this authentication process is called "two-way authentication".

To impose physical security, we use PUFs [32] in the proposed mechanism. PUFs can effectively protect UAVs from physical attacks because they provide each semiconductor device with

TABLE I
A COMPARISON OF RECENT RELATED WORKS

Item	Contributions	UAV-GS Authentication	UAV-UAV Authentication	Mutual Authentication	PUFs	Fuzzy Extractors	Intelligent Adaptation of Session Time
Pu <i>et al.</i> [14]	Mutually authenticates drones and zone service providers, with secure session keys and performs mutual authentication between drones	Yes	Yes	Yes	Yes	No	No
Zhang <i>et al.</i> [15]	Demonstrates that the elliptic curve cryptography (ECC)-based scheme for IoD authentication fails to prevent drone impersonation attacks	No	Yes	No	No	No	No
Hussain <i>et al.</i> [16]	Proposes an ECC-based authentication mechanism to ensure secure communication between a drone and a user	No	No	Yes	No	No	No
Feng <i>et al.</i> [17]	Designs a blockchain-enabled cross-domain authentication mechanism for IoD	No	Yes	Yes	No	No	No
Yu <i>et al.</i> [18]	Proposes an authentication scheme that uses PUF technology for IoD to introduce useful and reliable services in smart cities	Yes	No	Yes	Yes	No	No
Lounis <i>et al.</i> [19]	Designs an authentication protocol for inter-drone communication using PUF technology	No	Yes	Yes	Yes	No	No
Tanveer <i>et al.</i> [20]	Designs a mechanism for authenticated key management for IoD, along with the verification of the user's authenticity	Yes	No	Yes	No	No	No
Tan <i>et al.</i> [21]	Proposes a blockchain-based distributed authentication scheme for industrial UAVs in which the blockchain supports the immutable and distributed storage of UAVs' authentication information	Yes	Yes	Yes	No	No	No
Raja <i>et al.</i> [22]	Proposes a multi-layered blockchain-based 6G IoD ecosystem that addresses 6G-assisted IoD's authentication and trust issues using secure control mechanisms	Yes	No	Yes	No	No	No
Akram <i>et al.</i> [23]	Designs a deep learning-based key exchange mechanism for UAV-enabled vehicular ad-hoc networks (VANETs)	No	No	Yes	No	No	No
Alladi <i>et al.</i> [8]	Proposes a PUF-based mutual authentication mechanism for 5G-aided UAV networks	Yes	No	Yes	Yes	No	No
Alladi <i>et al.</i> [12]	Designs PUF-based UAV-GS and UAV-UAV mutual authentication protocols	Yes	Yes	Yes	Yes	No	No
Bansal <i>et al.</i> [11]	Designs a PUF-based BS-UAV mutual authentication scheme for UAV swarms	Yes	No	Yes	Yes	No	No
Pu <i>et al.</i> [13]	Achieves secure GS-UAV communications by designing a PUF-based mutual authentication protocol	Yes	No	Yes	Yes	No	No
Lei <i>et al.</i> [24]	Designs a security authentication protocol for IoD networks	Yes	No	No	Yes	No	No
Wazid <i>et al.</i> [25]	Proposes a scheme for authentication and key agreement between a drone and a user using a server	No	No	Yes	No	Yes	No
Our approach	Proposes PUF-based UAV-GS and UAV-UAV mutual authentication schemes with the dynamic session time adaptation	Yes	Yes	Yes	Yes	Yes	Yes

a unique identifier that is based on physical variations that occur naturally during a semiconductor's manufacturing process. PUFs are physical entities that are incorporated in physical structures, which makes them suitable for the high-security authentication of physical objects. An adversary cannot copy a UAV's PUF and clone it, which makes PUFs useful for ensuring physical-level authentication at UAVs.

Extending the proposed UAV-GS authentication, we design a UAV-UAV authentication mechanism that creates a secure session between two UAVs with the help of the GS. After the successful authentication of the UAVs and GS, the session time is dynamically set using a Thompson sampling (TS)-based multi-armed bandit (MAB) approach [33]. Therefore, overall UAAS is an intelligent mechanism for implementing a secure authenticated session between a UAV and the GS or two UAVs. The major contributions of this work are listed as follows:

- 1) We design a PUF-based authentication mechanism to mutually authenticate a UAV and the GS using a two-way authentication approach. Following this, a secure authenticated session is established between the UAV and GS, and a secure session key is generated for the communication between the two. To overcome the noise in PUFs, we use a *fuzzy extractor* [34] that works on the PUF outputs. Both UAVs and the GS do not store any secret information that can be used to decrypt messages communicated between them.
- 2) To maximize the busy time of active open sessions, we dynamically change the session time to minimize unnecessary idle time. Consequently, an adversary will have less time for an attack, thus enhancing the security of an authenticated session. To this end, based on the probability distribution of the UAVs' busy time, we use a TS-based online learning approach to intelligently adapt the session time of a secure session. In this context, busy time specifies the time when a communication takes place between two parties.
- 3) Extending the proposed UAV-GS authentication scheme, we design an authentication mechanism that authenticates two UAVs with the help of the GS and accordingly establishes a secure session for further communication between the UAVs. A secure session key is also generated for mutual communication between the UAVs.
- 4) We present a formal analysis of the security performance of UAAS using Mao and Boyd's well-known logic [35], the real-or-random (ROR) model [36], and the Scyther tool [37]. In addition, we also provide an informal security proof of UAAS by analyzing the security and versatility considering different mainstream attacks.
- 5) We present a comparative study of UAAS and related existing mechanisms in terms of different security features, and computation and communication overheads.
- 6) In order to analyze the practical impact of UAAS, we implement it in network simulator NS-3.35 [38], where we study different network performance metrics, such as throughput, end-to-end delay, and packet loss rate (PLR). The results show that UAAS significantly improves the network performance compared to baseline schemes.

C. Organization of This Paper

The organization of this paper is as follows. Section II presents the background and the proposed system model. Section III describes the UAV-GS and UAV-UAV authentication mechanisms along with the UAV registration scheme. Details of the dynamic adaptation of the session time are discussed in Section IV. In Section V, we present a thorough security analysis of UAAS through both formal and informal security analyses. Then, in Section VI, we present a comparative study of UAAS and related existing mechanisms. Section VII describes the practical impact of UAAS through simulation. Finally, we draw the conclusion in Section VIII.

II. UAAS: BACKGROUND AND SYSTEM MODEL

In this section, we provide a description on PUFs and fuzzy extractors before presenting the proposed system model with security goals.

A. Physically Unclonable Functions

PUFs are designed to uniquely map an input to an output based on the physical variations of the microstructure of a semiconductor device. An input-output pair is commonly known as a pair of *challenge-response* that is unique to each PUF circuit [39].

Definition 1: Let PUF_1 be a PUF value, and R_a and R_b be the output responses of the challenges $C_a \in \{0, 1\}^k$ and $C_b \in \{0, 1\}^k$, respectively. PUF_1 is secure if R_a and R_b have at least d variations, and for two PUFs, PUF_1 and PUF_2 , challenge C_a produces distinct responses $R_a, R_b \in \{0, 1\}^k$ with at least d' variations. Therefore, we have

$$\Pr[HD(PUF_1(C_a), PUF_1(C_b)) > d] = 1 - \epsilon,$$

$$\Pr[HD(PUF_1(C_a), PUF_2(C_a)) > d'] = 1 - \epsilon,$$

where HD is the Hamming distance and ϵ is a small value, challenges C_a and C_b are selected randomly, and d and d' denote the error tolerance thresholds for PUF_1 and PUF_2 , respectively.

If we realize the concept of the PUF using the cryptographically secure pseudorandom number generator (CSPRNG), we need to store the secure key used by the CSPRNG in a tamper-proof module (TPM) in the UAV. Therefore, it is required to design an additional module, TPM, along with the design of a UAV authentication protocol. Moreover, the CSPRNG is based on random number generations, which use entropy generally obtained from the operating system. In practice, the required number of random numbers is higher than the available entropy, and also, the extraction of random numbers from a system is slow. In addition, since the CSPRNG takes more than one input argument and depends on the random number generation from the operating system, the computation of the CSPRNG is slow compared to that of the PUF. Thus, we use the PUF in our proposed mechanism.

B. Fuzzy Extractor

Although PUF circuits can provide high reliability, the noise in PUFs remains an issue. To overcome this challenge, the

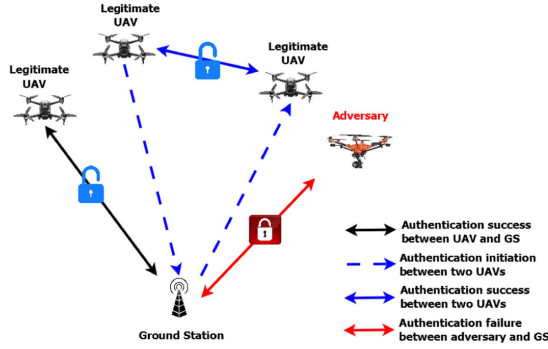


Fig. 1. System model

authors in [40] resorted to a fuzzy extractor [34]. The fuzzy extractor uses two methods – (i) probabilistic generation function denoted by $Gen(\cdot)$ and (ii) deterministic re-production function denoted by $Rep(\cdot)$. These two functions are defined in what follows.

- *Gen*: For challenge C_i , let the response be Res_i , i.e., $PUF(C_i) = Res_i$. $Gen()$ outputs a tuple (R_i, hd_i) , i.e., $Gen(Res_i) = (R_i, hd_i)$, where R_i and hd_i are the secret key and helper data, respectively.
- *Rep*: Let Res'_i be a PUF output. $Rep()$ can determine the original R_i based on hd_i , considering the Hamming distance between Res_i and Res'_i does not exceed an error tolerance threshold ρ . Therefore, $Rep(Res'_i, hd_i) = R_i$.

Considering the error tolerance threshold discussed in [41], if the Hamming distance between Res_i and Res'_i is λ and n_b is the number of input bits, we have $\rho = \lambda/n_b$. Next, we discuss the system model considered in this work.

C. System Model

As shown in Fig. 1, the considered system is composed of a ground station (GS), legitimate UAVs, and an adversary, where multiple UAVs can be connected to the GS simultaneously. In general, UAAS establishes an authenticated communication session between the GS G and a UAV U_i using a mutual authentication approach. The mutual authentication scheme is further extended to establish a secure session between two UAVs. For instance, we consider two UAVs U and V , where UAV U wants to make securely connect to V . In this context, first U establishes a secure session between U and G . Following this session, U requests G to create an authenticated session with V . Then, V and G are mutually authenticated as it was accomplished between U and G . In the aforementioned authentication scenarios, the duration of the secure session is adjusted using the TS-based learning mechanism proposed in UAAS.

When a UAV is deployed, it is registered with the GS using a secure channel. A PUF is associated with each UAV, where the PUF generates a response for a challenge that is randomly generated by the UAV. Let the challenge-response pair of U 's PUF be (C_u, R_u) . When U successfully authenticates with G , a separate challenge-response pair (C'_u, R'_u) , used to replace the existing (C_u, R_u) pair, is generated. Let the permanent unique identifier (UID) of U be UID_u and the UID of G be $GSID$. In addition, a temporary dynamic ID, known as DID_u , is generated

TABLE II
LIST OF PRIMARY NOTATIONS

Notation	Description
DID_u	Dynamic (temporary) ID of U
UID_u	Permanent UID of U
$GSID$	Permanent UID of G
(C_u, R_u)	PUF challenge-response pair of U
$h(\cdot)$	Hash function
SK_u	Session key for secure communication between U and G
$SK_{u,v}$	Session key for secure communication between U and V
N_u	Random nonce generated by U
$\oplus$	XOR operation
$\parallel$	Concatenation operation
ST	Session time
P^{busy}	Fraction of busy time of a session
$\mathcal{L}$	Number of association of a UAV with the GS

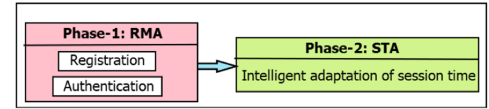


Fig. 2. Two-phase design of UAAS

by U when it registers or performs a mutual authentication with the GS. DID_u helps create a pseudo identity for U in each new authenticated session such that UID_u is not revealed to an adversary. In UAAS, DID_u is split into three parts and the splitting mechanism is known to both UAV_u and the GS. Table II lists the primary notations used in UAAS.

UAVs are vulnerable to a variety of attacks, including masquerade, man-in-the-middle, denial-of-service (DoS), and replay attacks. As a result, there is always a risk that an adversary will attempt to breach a UAV's security during an authenticated session between it and the GS, or between it and another UAV. When a session is idle, there is no data being communicated between the two authenticated parties. However, this period may provide an adversary with additional time to attack the secure session and obtain the session key. Therefore, it is important to minimize the amount of idle time in an authenticated session to reduce the risk of an attack.

D. Adversarial Model

An adversary $\mathcal{A}$ may try to modify the messages transmitted in a network, or eavesdrop or perform replay attacks on them. In addition, $\mathcal{A}$ may masquerade as a legitimate UAV and try to establish an authenticated session with the GS. $\mathcal{A}$ may also alter the messages being communicated between two parties using a man-in-the-middle attack. Moreover, a cloning attack could also be performed, and a node capturing attack could help $\mathcal{A}$ to extract information from the UAVs' memory.

E. Two-Phase Design of UAAS

Since we perform UAV-GS and UAV-UAV authentications before dynamically adapting the session time of the established secure session, we use a *two-phase* design approach, shown in Fig. 2. These two phases are described in what follows.

- 1) *Phase 1 - Registration and Mutual Authentication (RMA)*: This phase has two functionalities - (i) UAV registration and (ii) UAV authentication. In UAV registration, a UAV is registered with the GS, while in UAV authentication, a

communication session, between a UAV and the GS or two UAVs, is authenticated. A secure session establishment between two UAVs is based on the successful mutual authentication between the UAV and GS.

- 2) *Phase 2 - Session Time Adaptation (STA)*: In the STA phase, the session time of an authenticated session is dynamically adapted using a TS-based approach. Thus, both UAV-GS and UAV-UAV authentications are followed by the STA phase.

The RMA phase is run at both the GS and UAVs. The STA phase is executed only by the GS since it sets the session time after a secure session is established. Next, we discuss the security goals of UAAS.

F. Security Goals

In accordance with the work in [12], the security goals of UAAS are as follows.

- 1) For each authenticated session, the mechanism must generate a DID and a unique session key.
- 2) Authentication related important parameters, such as R and nonce, are communicated between a UAV and the GS in encrypted forms.
- 3) To initiate a session, mutual authentication must be achieved between a legitimate UAV and the GS.
- 4) UAAS must provide security against common threats, such as man-in-the-middle, masquerade, and replay attacks.
- 5) UAAS must protect the system from cloning attacks and node tampering.
- 6) No unauthorized party should be able to track the temporary ID of a UAV

Next, we present details of UAAS.

III. UAV REGISTRATION AND MUTUAL AUTHENTICATION

In this section, we describe the UAV-GS and UAV-UAV authentication approaches in UAAS. The UAV-GS authentication is performed over two phases – (i) UAV registration and (ii) UAV-GS authentication. During these phases, DID_u is split into three parts - P_1 , P_2 , and P_3 . The splitting mechanism is only known at all legitimate UAVs and the GS, where the logic behind the splitting mechanism is not shared with any third party. Details of the aforementioned phases are provided in what follows.

A. UAV Registration

Each U must be registered with the G before the deployment. The registration process is completed through *two steps* which are described below.

Step-1 (Identity code generation at U):

- 1) U generates a challenge C_u randomly and applies the PUF and fuzzy extractor to produce Res_u as

$$Res_u = PUF(C_u) \text{ and } (R_u, hd_u) = Gen(Res_u).$$

- 2) DID_u is generated as $DID_u = h(UID_u, r)$, where r is a random number.

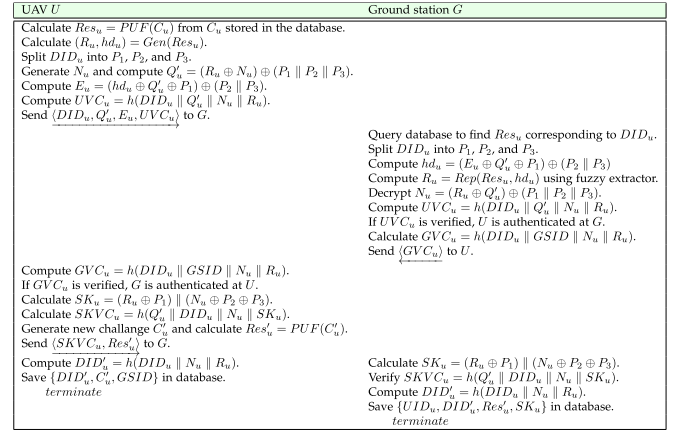


Fig. 3. UAV-GS authentication in UAAS

- 3) DID_u is split into P_1 , P_2 , and P_3 such that $DID_u = P_1 \parallel P_2 \oplus P_3$, where P_1 , P_2 , and P_3 are randomly generated bit strings. A parameter Q_u is generated as

$$Q_u = R_u \oplus P_1 \parallel P_2 \parallel P_3.$$

- 4) The *identity code (IC)* of U is computed as $IC_u = h(DID_u \parallel Q_u \parallel R_u)$.
- 5) U sends DID_u , Q_u , Res_u , and IC_u to the GS, and stores $\{DID_u, C_u, GSID\}$ in its database.

Step-2 (Identity code verification at G):

- 1) When G receives DID_u from U , it splits DID_u into P_1 , P_2 , and P_3 , following the procedure used by U .
- 2) R_u is decrypted from Q_u using the following operation

$$R_u = Q_u \oplus P_1 \parallel P_2 \parallel P_3.$$

- 3) Based on DID_u , Q_u , and R_u , the value of IC_u is computed to verify the corresponding value received from U .
- 4) The tuple $\{UID_u, DID_u, Res_u\}$ is stored in G 's database.

B. UAV-GS Authentication

UAAS achieves a mutual authentication between the GS and a UAV, and then a secure key is generated for a communication session between G and U , where further communication is carried out using that key. Therefore, UAAS ensures that a secure session is established between an authenticated GS and a legitimate UAV before the data transmission session starts. Fig. 3 shows the different messages which are exchanged in UAAS and details of the various computation are described below.

Step-1 (UAV verification code generation at U):

- 1) When U wants to establish a secure session with G , it applies the PUF and generates response Res_u using the challenge C_u stored in its database.
- 2) U generates R_u from Res_u as $R_u = Rep(Res_u, hd_u)$, where hd_u is helper data of U .
- 3) DID_u is split into P_1 , P_2 , and P_3 , following the approach adopted in the registration phase.

- 4) A random nonce, N_u , is generated and used to compute an updated version of Q_u as

$$Q'_u = (R_u \oplus N_u) \oplus (P_1 \parallel P_2 \parallel P_3).$$

- 5) To encrypt hd_u , a value of E_u is computed as $E_u = (hd_u \oplus Q'_u \oplus P_1) \oplus (P_2 \parallel P_3)$.
- 6) Based on N_u , U computes a UAV verification code (UVC), UVC_u , as $UVC_u = h(DID_u \parallel Q'_u \parallel N_u \parallel R_u)$. The UVC is used to verify a UAV at the GS.
- 7) U sends DID_u , Q'_u , E_u , and UVC_u to G .

Step-2 (UAV verification at G and GS verification code generation):

- 1) When G receives the information sent by U , it queries its database to find the Res_u value that corresponds to DID_u . If Res_u is found, the GS splits DID_u into P_1 , P_2 , and P_3 following the approach adopted in the registration phase and then computes $hd_u = (E_u \oplus Q'_u \oplus P_1) \oplus (P_2 \parallel P_3)$.
- 2) R_u is computed as $R_u = Rep(Res_u, hd_u)$ using the fuzzy extractor's reproduction function.
- 3) Based on Q'_u , the nonce N_u is decrypted as

$$N_u = (R_u \oplus Q'_u) \oplus (P_1 \parallel P_2 \parallel P_3).$$

- 4) From DID_u , Q'_u , N_u , and R_u , the value of UVC_u is calculated using $UVC_u = h(DID_u \parallel Q'_u \parallel N_u \parallel R_u)$. If UVC_u is verified with the UVC received from U , the GS successfully authenticates U .
- 5) To verify the GS at U , the GS calculates GS verification code (GVC), GVC_u , as

$$GVC_u = h(DID_u \parallel GSID \parallel N_u \parallel R_u)$$

- 6) G sends GVC_u to U .

R_u is one of the parameters that is required to compute UVC_u , which is used to verify the UAV, and R_u is calculated at the GS as $R_u = Rep(Res_u, hd_u)$. Thus, both Res_u and hd_u are required to compute R_u . Res_u is stored in G 's database, however hd_u is shared in an encrypted form of E_u as $E_u = (hd_u \oplus Q'_u \oplus P_1) \oplus (P_2 \parallel P_3)$. Therefore, the authentication process remains secure even if Res_u is revealed.

Step-3 (GS verification and session key generation at U):

- 1) U computes GVC_u based on the $GSID$ stored in its database. If GVC_u is verified with the GVC sent by G , it is verified at U .
- 2) To encrypt messages in further communication, the session key SK_u is computed as

$$SK_u = (R_u \oplus P_1) \parallel (N_u \oplus P_2 \oplus P_3).$$

- 3) U calculates the session key verification code (SKVC) as $SKVC_u = h(Q'_u \parallel DID_u \parallel N_u \parallel SK_u)$.
- 4) U generates a new challenge C'_u and calculates $Res'_u = PUF(C'_u)$. The $Gen()$ function is applied on Res'_u to generate a new response R'_u and helper data hd'_u .
- 5) U sends $SKVC_u$ and hd'_u to G .

Step-4 (Session key verification at G):

- 1) The session key SK_u is computed as $SK_u = (R_u \oplus P_1) \parallel (N_u \oplus P_2 \oplus P_3)$.
- 2) $SKVC_u$ is verified by calculating $h(Q'_u \parallel DID_u \parallel N_u \parallel SK_u)$. Thus, the same SK_u is generated at both U and G .

Therefore, mutual authentication is achieved between U and G and the session key is verified at both U and G . Both the GS and UAV compute the new DID'_u and update it as

$$DID'_u = h(DID_u \parallel N_u \parallel R_u).$$

The databases are updated at U and G to $\{DID'_u, C'_u, GSID\}$ and $\{UID_u, DID'_u, Res'_u, SK_u\}$, respectively. When a session between U and G becomes inactive, SK_u is deleted from G 's database.

Even if the adversary manages to learn the splitting mechanism used by DID_u , it will not be sufficient to discover N_u , as decrypting the latter requires R_u . Computing R_u requires Res_u , which is not stored in the UAV's database or transmitted to the GS when authenticating the UAV. If N_u is not decrypted, the adversary will not be able to compute UVC_u and, as a result, will not be authenticated at the GS.

How Is the Authentication Process Secure?

The environment is open; however, the adversary cannot deduce R_u because DID_u is split into P_1 , P_2 , and P_3 such that $DID_u = P_1 \parallel P_2 \oplus P_3$, and this splitting mechanism is known to only U and the GS. Thus, R_u cannot be intercepted through $R_u = Q_u \oplus DID_u$. Since the adversary cannot deduce R_u , it cannot compute IC_u , which is calculated as $IC_u = h(DID_u \parallel Q_u \parallel R_u)$.

In the subsequent authentication process, UVC_u is computed as $UVC_u = h(DID_u \parallel Q'_u \parallel N_u \parallel R_u)$, where N_u and R_u are required and not transmitted by U to the GS. N_u is a random nonce that is generated by U and decrypted at the GS as $N_u = (R_u \oplus Q'_u) \oplus (P_1 \parallel P_2 \parallel P_3)$, where R_u , P_1 , P_2 , and P_3 are computed by U and the GS individually. R_u is calculated at the GS from Res_u and hd_u using the fuzzy extractor's $Rep(\cdot)$. Computing R_u requires hd_u , which is shared in an encrypted form of E_u as $E_u = (hd_u \oplus Q'_u \oplus P_1) \oplus (P_2 \parallel P_3)$. Therefore, N_u and R_u cannot be intercepted by the adversary, and consequently, UVC_u can be deduced by only the GS and U . As a result, the adversary cannot be authenticated by the GS. Moreover, since R_u and N_u are used to compute GVC_u , the GS can be authenticated by a legitimate U . Therefore, the authentication process is secure at both U and the GS.

C. UAV-UAV Authentication

Considering the UAV-GS authentication mechanism discussed in Section III-B, a secure session can be established between two UAVs with the help of the GS. Assume that U wants to securely connect with V ; therefore, U needs to know the unique ID, UID_v , of V . Details of the UAV-UAV authentication steps are presented in Fig. 4 and described in what follows.

Step-1 (at U):

- 1) U establishes a secure session with G using the UAV-GS authentication mechanism discussed in Section III-B.
- 2) Let Req_u be the authentication request string. U computes $h(Req_u \parallel DID_u \parallel UID_v \parallel SK_u)$ and sends it to G with Req_u , DID_u , and UID_v .

Step-2 (at G):

- 1) G queries its database to find any entry for the session key SK_u against DID_u received from U . If SK_u is found, a

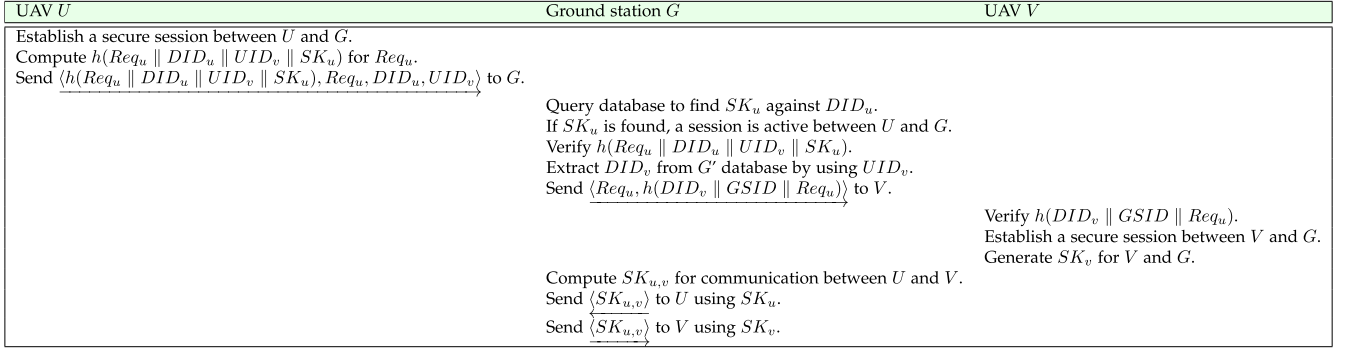


Fig. 4. UAV-UAV authentication in UAAS

session is active between U and G ; otherwise, G informs U to create a new session.

- 2) If there is an active session between U and G , $h(Req_u \parallel DID_u \parallel UID_v \parallel SK_u)$ is verified based on SK_u , which is found in G 's database. If the hash is verified, the dynamic ID, DID_v , of V is extracted from G 's database using UID_v .
- 3) G sends Req_u and $h(DID_v \parallel GSID \parallel Req_u)$ to V .

Step-3 (at V):

- 1) V verifies $h(DID_v \parallel GSID \parallel Req_u)$ against the values of DID_v and $GSID$ stored in V 's database.
- 2) If the hash is verified, a secure session is established between V and G using the UAV-GS authentication mechanism discussed in Section III-B. Let SK_v be the session key between V and G .

Step-4 (at G): For the communication between U and V , G computes a session key $SK_{u,v}$ which is distributed to U and V using the previously generated secure session keys SK_u and SK_v , respectively. Therefore, based on $SK_{u,v}$, U and V can securely communicate with each other.

Next, we discuss the session duration management using TS.

IV. THOMPSON SAMPLING-BASED SESSION TIME ADAPTATION

When UAV-GS or UAV-UAV authentication is completed, a secure communication session is established between the authenticated parties. An authenticated session has a set session time during which the two parties can communicate with each other without the need for further authentication. When the session time expires, the session becomes inactive and the two parties need to complete the authentication procedure again if they want to continue communicating with one another. UAAS adopts the TS [33]-based RL method to dynamically determine how long the session time should be for a successfully authenticated session. A description of TS is given below.

A. Thompson Sampling

TS is based on Bayesian inference and applies an online heuristic approach that has the objective to maximize the reward by choosing actions considering randomly drawn beliefs. A probabilistic distribution is used by engaging a reward associated with an *action* (arm). Using TS, the learner tries to maximize

the expected reward based on past knowledge (*exploitation*) and gain knowledge about a new environment (*exploration*).

Let the sets of contexts, actions, and rewards be represented by $\mathcal{Y}$, $\mathcal{A}$, and $\mathcal{R}$, respectively. The learner obtains a *reward* $r \in \mathcal{R}$ each time it takes an action $a \in \mathcal{A}$ in context $y \in \mathcal{Y}$. The expected reward's distribution depends on the current context and action. The learner determines the actions that maximize the cumulative expected rewards. Let the probability distribution of r and past observation triplet be represented by θ and $\mathcal{F}$, respectively, where $\mathcal{F} = \{(y; a; r)\}$. Let $P(\theta)$ be the prior probability distribution on θ . Given θ , a , and y , let $P(r|\theta, a, x)$ denote the likelihood of the reward r . For θ , let the posterior probability distribution be $P(\theta|\mathcal{F}) \propto P(\mathcal{F}|\theta)P(\theta)$, where $P(\mathcal{F}|\theta)$ is the likelihood. In TS, action $a^* \in \mathcal{A}$ is played by the learner following the probability distribution that maximizes the cumulative expected reward, and that probability is defined as

$$\int \mathbb{I} \left[\mathbb{E}(r|\theta, a^*, y) = \max_a \mathbb{E}(r|\theta, a', y) \right] P(\theta|\mathcal{F}) d\theta, \quad (1)$$

where $\mathbb{I}$ and $\mathbb{E}$ denote the *indicator function* and expected value, respectively.

B. Reasons for Applying TS in Session Time Adaptation

The probability distribution is suitable for quantifying uncertainty. The prior can determine the probability distribution of an event without considering any advance knowledge of it. The probability distribution of an uncertain event can be determined based on the likelihood computed by TS by considering its past performance. TS computes the probability of exploitation considering a well-balanced mix of exploitation and exploration using the likelihood, prior, and posterior distributions. As a result, TS is able to dynamically adjust in an unknown environment. Since the session time depends on the duration of the UAV's active communication with the GS, it cannot be estimated directly from past communication information. Therefore, in a time-varying wireless environment, TS can help adapt the session time dynamically based on the probabilistic approach considering the past and predicted future impacts of the session time on the maximization of a UAV's busy time. Other reinforcement learning algorithms do not use a probabilistic approach to quantify an event in advance or consider the past

performance of an event. In the case of supervised learning, a dataset is required for initial model training, and the model must be retrained regularly to keep it up to date.

C. Application of TS in UAAS

In UAAS, the action a specifies the session time. At any instance during an active session, let θ represent the fraction of time a UAV is busy. In our model, a statistical table, denoted by $\mathcal{E}$, is used to keep information on past UAV connections. We define $\mathcal{E}$ as $\mathcal{E} = \{\mathcal{U}, ST, P^{busy}, \mathcal{L}\}$, where ST and P^{busy} are the session time and the fraction of busy time of the UAV with unique ID $UID \in \mathcal{U}$, respectively, where $\mathcal{U}$ is set of all UIDs. Moreover, $\mathcal{L}$ is the number of associations of a UAV with the GS, i.e., $\mathcal{L}$ counts the number of times the UAV connected to the GS. In what follows, we provide a description of the different TS parameters in the context of our application.

- **Prior distribution:** Considering the probability of busy time of a UAV, the prior is determined for the next session time. Since our objective is to minimize the idle time of a connected UAV, we consider the average of the fraction of busy time as the prior of the corresponding session time. Let P_u^{busy} be the fraction of busy time of U . Thus, if $\hat{P}_u^{busy}$ denotes the average of P_u^{busy} , following the description of TS, the prior can be represented as $P(\theta) = \hat{P}_u^{busy}$, which can be obtained using statistical table $\mathcal{E}$. Therefore, the prior specifies the probability of selecting a session time. $P(\theta)$ is calculated for each distinct U in $\mathcal{E}$.
- **Likelihood:** The fraction of busy time can be influenced by the number of association $\mathcal{L}$. Thus, in UAAS, we represent the likelihood by $P(\mathcal{L}|\theta)$. In particular, the likelihood defines the probability of $\mathcal{L}$ for a given average of fraction of busy time. For U , if $\mathcal{L}_u^{sum}$ denotes the total number of associations under $[P_u^{busy} - \delta, P_u^{busy} + \delta]$, for small $\delta > 0$, we define $P(\mathcal{L}|\theta) = \frac{\mathcal{L}_u^{sum}}{\mathcal{L}^{sum}}$, where $\mathcal{L}^{sum}$ is the sum of all associations. The likelihood is determined for each distinct U in $\mathcal{E}$.
- **Posterior distribution:** In UAAS, let $P(\theta|\mathcal{L})$ denote the posterior distribution. Considering the prior and likelihood, the posterior distribution finds the probability of choosing the best possible session time for an established secure session. Therefore, following TS, $P(\theta|\mathcal{L})$ is calculated as

$$\begin{aligned} P(\theta|\mathcal{L}) &\propto P(\mathcal{L}|\theta) \times P(\theta) \\ \Rightarrow P(\theta|\mathcal{L}) &= \lambda P(\mathcal{L}|\theta) \times P(\theta). \end{aligned} \quad (2)$$

Here, λ is a constant of proportionality. We consider λ as the number of distinct UAV entries in $\mathcal{E}$. The posterior distribution is calculated for all the computed prior and likelihood values. Afterwards, the session time, for which the posterior distribution is maximized, is applied for the newly authenticated session.

Algorithm 1 presents the description of UAAS. Assume that U with UID_u has requested for an authentication and ST_u is the corresponding session time. To initialize the statistical table $\mathcal{E}$ and explicitly explore the session time, we use the ϵ -greedy policy discussed next.

Algorithm 1: Session Time Adaptation in UAAS.

```

1: Input:  $\mathcal{E}$  and  $UID_u$ .
2: Output:  $ST_u$ .
3: Initialization: Let  $t_d$  be the initial time for the
   exploration of session time and populating the statistical
   table  $\mathcal{E}$ . The session time is chosen randomly and after
   the completion of a session,  $\mathcal{E}$  is updated accordingly.
4: while  $t > t_d$  do
5:   Calculate  $\epsilon_t$  by using (3).
6:   Let  $\xi \leftarrow \text{Random}(0,1)$ .
7:   if  $\xi \leq \epsilon_t$  then
8:     if  $UID_u \in \mathcal{U}$  in  $\mathcal{E}$  then
9:       Calculate  $P(\theta)$ ,  $P(\mathcal{L}|\theta)$ , and then compute
        $P(\theta|\mathcal{L})$  by following (2).
10:      Select the session time that maximizes  $P(\theta|\mathcal{L})$ .
11:    else
12:      Choose a session time uniformly at random in
      statistical table  $\mathcal{E}$ .
13:    end if
14:    Select a session time randomly.
15:  end if
16:  Update  $\mathcal{L}$ .
17: end while
18: Return  $ST_u$ .
19: End

```

D. Exploration-Exploitation Using ϵ -greedy Policy

The ϵ -greedy [42] mechanism maintains a balance between exploration and exploitation. At time t , the exploration probability, ϵ_t , is defined as

$$\epsilon_t = \min(1, eB/k^2t), \quad (3)$$

where B denotes the total number of UAVs connected to the GS. The parameter $e \geq 0$ is a small integer. The parameter k is the difference between the maximum and minimum values of P^{busy} .

The ϵ -greedy policy is described by two phases – (i) exploration, where an action is chosen randomly and (ii) exploitation, where the action that maximizes a given objective is selected. In our case, the objective is to maximize the posterior distribution $P(\theta|\mathcal{L})$. As mentioned in [43], after n runs, the probability of applying a suboptimal action is upper bounded by $O(e/k^2n)$, where $n \geq eB/k^2$.

E. Time Bound Analysis of TS-Based Session Time Adaptation

In the proposed TS-based mechanism, the prior is the average busy time, i.e., $P(\theta) = \hat{P}_u^{busy}$ and $P(\theta)$ is calculated for each distinct U in $\mathcal{E}$. Thus, if M denotes the number of distinct UAVs in $\mathcal{E}$, it is required to search $\mathcal{E}$ to find M . Therefore, the search time for M is bounded by $\mathcal{O}(|\mathcal{E}|)$, where $|\mathcal{E}|$ defines the size of $\mathcal{E}$. Considering distinct entries of UAVs in $\mathcal{E}$, the likelihood is calculated in terms of the probability of $\mathcal{L}$ for a busy time fraction. Thus, the computation of the likelihood has the time bound of $\mathcal{O}(|\mathcal{E}|)$. As the posterior distribution is computed for all

the computed prior and likelihood values, the time complexity of the posterior distribution computation is influenced by the number of distinct UAV entries in $\mathcal{E}$, and thus the computation of the posterior distribution is bounded by $\mathcal{O}(M)$. Therefore, the sum of the time bounds of calculating the prior, likelihood, and posterior distributions is $\mathcal{O}(|\mathcal{E}| + M) = \mathcal{O}(|\mathcal{E}|)$, since $M \subseteq |\mathcal{E}|$. Assume that $\mathcal{E}$ is sorted. Hence, the time bound of the TS-based scheme is reduced to $\mathcal{O}(\log |\mathcal{E}|)$.

Moreover, we also find the time complexity bound of both the exploration and exploitation approaches used in UAAS. In the exploitation, the maximum time will be required if U that requested a connection is available in $\mathcal{E}$. In this context, TS will be applied to find the best possible session time, and thus the time complexity of the exploitation is $\mathcal{O}(\log |\mathcal{E}| + M)$. However, if $U \notin \mathcal{E}$, a session time is chosen randomly in $\mathcal{E}$. Meanwhile, since the session time in the exploration is selected randomly, its time complexity does not depend on $\mathcal{E}$. Let us consider that the time complexity bound of ST adaptation is $T(ST \text{ adaptation})$. Hence, considering the TS-based scheme, the time complexity bound of the dynamic session time adaptation can be defined as

$$T(ST \text{ adaptation}) = \mathcal{O}(\log |\mathcal{E}|) \quad (4)$$

V. SECURITY ANALYSIS

We analyze the security of UAAS using both a formal security proof and an informal verification since the formal proof is insufficient on its own. For the formal security analysis, we use Mao and Boyd's well-known logic [35], the ROR model [36], and the Scyther tool [37]. The informal proof is based on the old-fashioned cryptanalysis, which is necessary because it can guarantee both versatility and security. While only the formal proof is insufficient [44], [45]. In UAAS, the UAV-UAV authentication is based on two instances of the proposed UAV-GS authentication, and therefore the proof for UAV-GS authentication is trivial.

A. Formal Security Analysis Through Mao and Boyd Logic

In the formal security analysis, we utilize the authentication, confidentiality, super-principal, nonce-verification, good-key, and fresh inference rules of Mao and Boyd logic [35]. First, we prove the statement " G believes N_u is a good secret key between U and G ". To this end, the statements in the Mao Boyd logic are represented in (5)–(14), and the discussion on the formal security proof is presented below.

Claim 1: G believes N_u is a good secret key between U and G .

Proof: R_u is transmitted in an encrypted form and G computes R_u using hd , stored in its database. Therefore, " G believes R_u is a good secret key between U and G ", i.e.,

$$G \models U \xrightarrow{R_u} G. \quad (5)$$

G can decipher N_u using R_u , and thus

$$G \xrightarrow{R_u} N_u. \quad (6)$$

We apply the *authentication rule* to (5) and (6) and we obtain the statement " G believes U encrypted N_u using R_u ", i.e.,

$$G \models U \xrightarrow{R_u} N_u. \quad (7)$$

Since U creates a new nonce N_u in each authentication process, " G believes N_u is fresh", and thus

$$G \models \#(N_u). \quad (8)$$

Applying the *nonce-verification rule* to (7) and (8), we get " G believes that U believes that R_u is a good secret key between U and G ", i.e.,

$$G \models U \models \xrightarrow{R_u} G. \quad (9)$$

Since U generates a new N_u in each time of the authentication, G considers that no one other than U could have seen N_u . Therefore, " G believes that U believes that none other than G can access N_u ", i.e.,

$$G \models U \models \{G\}^c \triangleleft N_u. \quad (10)$$

Now, we apply the *confidentiality rule* to statements (7), (9), and (10) and we get " G believes that U believes that none other than U and G can access N_u ". That is, we have

$$G \models U \models \{U, G\}^c \triangleleft N_u. \quad (11)$$

It is assumed that U is a legitimate UAV and G believes that U is trusted. This is represented as

$$G \models \text{sup}(U). \quad (12)$$

We apply the *super-principal rule* to (11) and (12) and we obtain " G believes that none other than G and U can access N_u ", i.e.,

$$G \models \{U, G\}^c \triangleleft N_u. \quad (13)$$

Now, applying the *good-key rule* to (8) and (13), we prove " G believes N_u is a good secret key between U and G ", i.e.,

$$G \models U \xrightarrow{N_u} G. \quad (14)$$

□

Next, we prove the statement " U believes N_u is a good secret key between U and G ". For this proof, the statements in the Mao Boyd logic are represented in (15)–(20).

Claim 2: U believes N_u is a good secret key between U and G .

Proof: Since R_u is generated by the PUF and R_u is sent to G in encrypted form, we have the statement " U believes R_u is a good secret key between parties U and G " (15). Moreover, a new nonce N_u is generated by U in each authentication process, and N_u is only shared with G in encrypted form. Therefore, " U believes that none other than G can access N_u " (16). U uses R_u to encrypt N_u (17). We apply the *confidentiality rule* to (15), (16), and (17) and get " U believes that none other than U and G can access N_u " (18)

$$U \models U \xrightarrow{R_u} G. \quad (15)$$

$$U \models \{G\}^c \triangleleft N_u. \quad (16)$$

$$U \stackrel{R_u}{\rightsquigarrow} N_u. \quad (17)$$

$$U \models \{U, G\}^c \triangleleft \| N_u. \quad (18)$$

Since a new nonce N_u is generated in each time of the authentication, U believes that N_u is fresh (19). We apply the *good-key rule* to (18) and (19) and prove “ U believes N_u is a good secret key between U and G ” (20)

$$U \models \#(N_u). \quad (19)$$

$$U \models U \xleftrightarrow{N_u} G. \quad (20)$$

□

Hence, an adversary cannot access N_u . Similarly, it can also be proved that R_u cannot be accessed by an adversary. Therefore, if an adversary does not have access to N_u and R_u , it cannot decrypt the communicated data, thus maintaining the secrecy of N_u and R_u , regardless of the type of attack, such as a man-in-the-middle, masquerade attack, or replay attack.

The fuzzy extractor’s output, R_u , is calculated from the PUF output, and hd_u is required to compute R_u . hd_u is shared in an encrypted form. Therefore, the fuzzy extractor helps provide a security level between the computation of the PUF and hash functions. Moreover, from the formal security analysis through Mao and Boyd logic, it is proved that N_u and R_u cannot be accessed by an adversary. Therefore, although the security is reduced to hash security in the proposed mechanism, an adversary cannot compute the hash functions because N_u and R_u are used to generate the hash values.

B. Formal Security Analysis Using ROR Model

In this section, we describe the ROR model [36] and then utilize it to perform a formal security analysis of UAAS. The random oracle model (ROM) is an oracle that responds to each query with a random response. In practice, the ROM does not model real life. This is because, in the real world, no random functions exist, and usually a single hash function is used. Therefore, an adversary may not use a random function and may work with a single hash function. In that case, the ROM will fail to provide a proper security analysis. However, the ROR model can respond with both randomized and non-randomized values.

1) *ROR Model*: The ROR model incorporates the following components:

- Participants*: Let π_U^u and π_G^g denote the u^{th} and g^{th} instances corresponding to U and G , respectively.
- Partnering*: Two instances, π_U^u and π_G^g , are called partners if the following two conditions are satisfied simultaneously: i) the session key is common to U and G , and ii) the partial transcript of the messages exchanged between them is unique.
- Freshness*: The instances, π_U^u and π_G^g , are fresh, and the session key SK used by U and G cannot be revealed to an adversary.
- Adversary*: An adversary $\mathcal{A}$ has control over the channel and can execute the following queries:
 - Execute*(π^u, π^g): By executing this query, $\mathcal{A}$ is allowed to intercept all the messages transmitted between

U and G . Therefore, this query models eavesdropping attacks.

- Send*(π^g, m): By executing this query, $\mathcal{A}$ sends a message, m , to π^g and receives a response message as a reply from π^g . This query models an active attack.
- Reveal*(π^u): After $\mathcal{A}$ executes this query, the current session key computed by U (and its partner) is disclosed to it.
- Test*(π^u, π^g): This query is used to determine the semantic security of SK established between U and G in accordance with the indistinguishability property of the ROR model [36]. At first, an unbiased coin c is tossed by $\mathcal{A}$, and the outcome decides the *Test* query’s result. If SK is a fresh value, U or G produces SK when $c = 1$, a random number when $c = 0$, or a null value otherwise.

Semantic security of the session key: The *Test*() query can be performed repeatedly to π^u and π^g , and the result is saved into bit r . The adversary wins the game when $r = r'$, where r' is guessed randomly. For $\mathcal{A}$, let $Adv_{S, \mathcal{A}}^{SKE}(t) = |2 \cdot Pr[SUCCESS] - 1|$ be the advantage in cracking the semantic security, $\mathcal{S}$, of the session key agreement in time t , where *SUCCESS* denotes an instance of winning the game.

2) *Security Proof*: Theorem 1 presents the semantic security of UAAS.

Theorem 1: Let $\mathcal{A}$ follow a polynomial time adversary that runs against $\mathcal{S}$. Let q_p, q_h, q_s , and l be the number of PUFs, hash, *Send*() queries, and bits in the session key, respectively. Assume that $|PUF|$ and $|Hash|$ denote the range spaces of $PUF(\cdot)$ and $h(\cdot)$, respectively. Then, we have

$$Adv_{S, \mathcal{A}}^{SKE}(t) \leq \frac{q_h^2}{|Hash|} + \frac{q_p^2}{|PUF|} + \frac{2q_s}{2^l}. \quad (21)$$

Proof: We follow the proof presented in [46] to proof this theorem. $G0 - G4$ are five sequential games. Let $SUCCESS_i$ denote that $\mathcal{A}$ successfully guesses c in $G_j, j \in [0, 4]$. The games are discussed below.

- Game G0*: Game $G0$ represents an actual attack on $\mathcal{S}$. Since c is guessed in the initial stage of $G0$, it follows that

$$Adv_{S, \mathcal{A}}^{SKE}(t) = |2 \cdot Pr[SUCCESS_0] - 1|. \quad (22)$$

- Game G1*: In game $G1$, an eavesdropping attack is detected in which $\mathcal{A}$ queries *Execute*() to intercept the messages transmitted between U and G . After that, $\mathcal{A}$ can query *Test*() to determine whether the actual SK or a random number is obtained. The adversary requires R_u, N_u , and the decomposition technique of DID_u to calculate SK . Since these values are not available to $\mathcal{A}$, only U and G can calculate SK . Therefore, the probability of $\mathcal{A}$ winning $G1$ is not any higher. Thus, we have

$$Pr[SUCCESS_1] = Pr[SUCCESS_0]. \quad (23)$$

- Game G2*: Using the hash and *Send*() queries, a legitimate participant can be convinced to accept a modified message. However, in the proposed mechanism, if the transmitted messages are modified, UVC_u and GVC_u will not be verified at the legitimate participants. Moreover, the hash

operations are dynamic and do not include the same variable set. Thus, based on the birthday paradox, we have

$$|Pr[SUCCESS_2] - Pr[SUCCESS_1]| \leq \frac{q_h^2}{2|Hash|}. \quad (24)$$

- *Game G3*: Game $G3$ is an extension of $G2$ that incorporates the PUF queries. Based on the analogous argument given in $G2$, the $PUF(\cdot)$ provides the result as

$$|Pr[SUCCESS_3] - Pr[SUCCESS_2]| \leq \frac{q_p^2}{2|PUF|}. \quad (25)$$

- *Game G4*: In this game, $\mathcal{A}$ tries to intercept the messages using the $Send()$ query to compute SK , and its probability of estimating the session key of l bits is $\frac{1}{2^l}$. In the absence of guessing attacks, $G3$ and $G4$ are identical. Therefore, we have

$$|Pr[SUCCESS_4] - Pr[SUCCESS_3]| \leq \frac{q_s}{2^l}. \quad (26)$$

Finally, to win $G4$, b' needs to be guessed after querying $Test()$. Thus, we have

$$Pr[SUCCESS_4] = \frac{1}{2}. \quad (27)$$

From (22), (23), (24), (25), (26), and (27), we have

$$\begin{aligned} Adv_{S,A}^{SKE}(t) &= 2|Pr[SUCCESS_0] - \frac{1}{2}| \\ &\leq 2|Pr[SUCCESS_1] - Pr[SUCCESS_2]| \\ &\quad + 2|Pr[SUCCESS_2] - Pr[SUCCESS_3]| \\ &\quad + 2|Pr[SUCCESS_3] - Pr[SUCCESS_4]| \\ &\leq \frac{q_h^2}{|Hash|} + \frac{q_p^2}{|PUF|} + \frac{2q_s}{2^l}. \end{aligned} \quad (28)$$

□

C. Formal Security Analysis Through Scyther

For formal security analysis, we also use Scyther [37], which is a broadly-accepted security analysis simulator. Scyther is a security protocol verification tool that verifies protocols with bounded and unbounded number of sessions and uses a symbolic backwards search technique based on patterns [47]. We implement UAAS in Scyther, where (i) the number of runs is set to 10, (ii) the matching type is set to “Find All Type Flaws”, (iii) the search pruning is set to “Find All Attacks”, and (iv) the maximum number of patterns per claim is set to “10”. In Scyther, “UAAS” is the name of the protocol, in which ‘U’ and ‘G’ are known as ‘roles’. There are 7 claims stated by each of the roles, which are shown in Fig. 5. It is noted that in UAAS, N_u and R_u are shared secret between U and G , which is mentioned in claim 1 and 2 (in both U and G). Claim 6 (*Alive*) establishes the liveness of all the roles throughout an execution of the protocol. If a role believes that it can interact with another role, the *Weakagree* claim is satisfied, which is given in claim 7. Claim 3 (*Nisynch*) defines the general consistency and synchronicity of the protocol, and additionally ensures that the event of receiving

Scyther results : verify

Claim	Status	Comments
UAAS U UAAS,i1 Secret Nu	Ok	No attacks within bounds.
UAAS,i2 Secret Ru	Ok	No attacks within bounds.
UAAS,i3 Nisynch	Ok	No attacks within bounds.
UAAS,i4 Niagree	Ok	No attacks within bounds.
UAAS,i5 Commit G,Nu,Ru	Ok	No attacks within bounds.
UAAS,i6 Alive	Ok	No attacks within bounds.
UAAS,i7 Weakagree	Ok	No attacks within bounds.
G UAAS,r1 Secret Nu	Ok	No attacks within bounds.
UAAS,r2 Secret Ru	Ok	No attacks within bounds.
UAAS,r3 Nisynch	Ok	No attacks within bounds.
UAAS,r4 Niagree	Ok	No attacks within bounds.
UAAS,r5 Commit U,Nu,Ru	Ok	No attacks within bounds.
UAAS,r6 Alive	Ok	No attacks within bounds.
UAAS,r7 Weakagree	Ok	No attacks within bounds.

Done.

Fig. 5. Simulation result under formal security analysis through Scyther

a message is preceded by the sending of the message. In claim 4, *Niagree* states that if a role acts as an initiator and performs one-way authentication with the other role, the initiator believes that when the protocol is completed, the data consistency is maintained with the other role acting as a responder. In claim 5, *Commit* states the successful execution of the roles on the secrets N_u and R_u .

D. Informal Security Analysis

In this subsection, through informal security proof, we analyze the security of UAAS considering different well-known attacks. Our analysis follows a list of criteria discussed in [48], which removes ambiguities and redundancies to proof the security features of a protocol. Several works, such as [12], [46], [49], have considered this type of security analysis technique. Since passwords and smart cards are not involved in our proposed mechanism, we have not considered the security features C1-C4, C6, and C9 in [48]. In addition, we also analyze the attainment of untraceability property and resistance to Ephemeral Secret Leakage (ESL) attacks. In particular, we analyze in our informal security analysis how the proposed mechanism meets the security goals presented in Section II-F. Our analysis of the attainment of forward secrecy (Section V-D7) and secure key agreement (Section V-D3) indicates that security goal 1 is achieved. Our analysis of resilience to common attacks (Section V-D1) and

the attainment of mutual authentication (Section V-D5) confirms that security goal 2 is achieved. Our analysis of the attainment of mutual authentication (Section V-D5) indicates that security goal 3 is achieved. Our analysis of resilience to common attacks (Section V-D1) confirms that security goals 4 and 5 are achieved. And finally, our analysis of the attainment of untraceability (Section V-D8) and user anonymity (Section V-D6) confirms that security goal 6 is achieved.

Detailed discussion on the informal security proof of UAAS is presented below.

1) *Resilience Against Common Attacks (C5)*: UAAS is secure against man-in-the-middle (MM) (C5.2), masquerade (C5.1), replay (C5.3), node tampering (C5.4), and cloning (C5.5) attacks. Since the PUF depends on the hardware of a device, adversary $\mathcal{A}$ cannot access the PUF of U and consequently cannot masquerade as U . Adversary $\mathcal{A}$ also cannot have the correct (DID_u, hd_u) pair, and thus it cannot masquerade as G . Moreover, N_u and R_u are transmitted in encrypted forms, where N_u and R_u are good secrets between U and G (proved in Section V-A). Therefore, masquerade and MM attacks will fail. $\mathcal{A}$ cannot replay old messages since a new (C_u, R_u) pair is applied for establishing each new authenticated session and the (C_u, R_u) pair is not stored in the database of either U or G . Thus, $\mathcal{A}$ does not have access to U 's (C_u, R_u) pair. In addition, in each authenticated session, a new DID_u is generated based on R_u and N_u . Along with R_u and N_u , DID_u is used to encode a message transmitted between U and G . Therefore, $\mathcal{A}$ cannot decode messages and perform replay attacks.

UAAS is resilient against node tampering attacks. This is because U 's PUF will be unusable if $\mathcal{A}$ physically captures U and attempts to tamper with U 's PUF. Despite the fact that a device capturing is occurred, $\mathcal{A}$ cannot access the (C_u, R_u) pair and other critical information such as N_u and R_u because this information is not stored in the database of either U or G . Since the PUF is inherently unclonable, adversary $\mathcal{A}$ cannot clone U and G . Moreover, during U 's registration, R_u is used and it is transmitted to G in an encoded message. Thus, UAAS is secure against cloning attacks.

2) *Resilience Against De-Synchronization Attacks (C5.6)*: After generating the session key SK_u , U sends $\{SKVC_u, Q''_u, hd'_u\}$ to G . If G does not receive it, it assumes that U has not received its last message, i.e., GVC_u , and thus it resends it. After a repeated number of attempts, if G does not get $\{SKVC_u, Q''_u, hd'_u\}$, G discards R_u and N_u . In addition, G does not update DID_u and terminates the authentication procedure for U because the two-way authentication is not completed. On the other side, if U does not receive GVC_u , it does not generate (C'_u, R'_u) , and thus the two-way authentication also fails. The last generated C_u and DID_u are kept by U , and the last updated DID_u and hd_u are stored in G 's database. Hence, UAAS overcomes de-synchronization and restores to the previous state when the present state cannot be completed.

3) *Secure Key Agreement (C7)*: The secrecy of nonce N_u is proved in Section V-A. Moreover, P_1 , P_2 , and P_3 are derived from DID_u , which is freshly computed based on N_u and R_u for each new secure session. Since N_u and R_u are new in each

session, a different DID_u is used to establish an authenticated session. Therefore, based on DID_u and R_u , a new session key is computed as $SK_u = (R'_u \oplus P_1) \parallel (N_u \oplus P_2 \oplus P_3)$, which is new for each successful authentication.

4) *Avoidance of Clock Synchronization (C8)*: Our proposed mechanism does not use timestamps. Therefore, clock synchronization and time delay do not need to be considered in UAAS, i.e., U does not synchronize its system clock with G and vice versa.

5) *Attainment of Mutual Authentication (C10)*: A new (C_u, R_u) is created in each authenticated session and R_u is not stored in either U 's or G 's database. Since R_u is transmitted in encrypted format between U and G , and R_u is used to generate DID_u , the session key SK_u can be established only between U and G . Thus, the establishment of a session key between two authenticating parties implies that the parties are legitimate and a mutual authentication is achieved between them.

The mutual authentication between U and G is also established based on UVC_u and GVC_u , and both of them are generated considering N_u and R_u . Thus, UVC_u and GVC_u are new in each authentication initiation and not accessible to adversary $\mathcal{A}$. If UVC_u is similar at both U and G , U is authenticated at G . Similarly, to authenticate G at U , GVC_u needs to be same at both U and G . Therefore, the mutual authentication is securely performed between U and G .

6) *Attainment of User Anonymity (C11)*: In UAAS, the dynamic ID, DID_u , of U is updated for each new authentication as $DID'_u = h(DID_u \parallel N_u \parallel R'_u)$, i.e., every new authentication procedure starts with a new temporary ID for the UAV, and therefore user anonymity is guaranteed in UAAS. Moreover, both U and G are able to calculate the same DID_u based on R_u and N_u , which are securely transmitted between U and G and not stored in their databases. Any other message exchange is also not required between them. Hence, only U and G can compute DID_u .

7) *Attainment of Forward Secrecy (C12)*: If $\mathcal{A}$ guesses the current session key SK_u , the security of future sessions will not be compromised. This is because in the next session, the (C_u, R_u) pair, N_u , and R_u will be different, and accordingly DID_u will be changed. As a result, the session key is different in each session. In addition, for U , the $SKVC_u$ must be verified at G to initiate the communication using SK_u . Moreover, in each authentication initiation, UVC_u and GVC_u are different based on N_u and R_u . For generating a session key, UVC_u and GVC_u should be verified first at G and U , respectively. Thus, UVC_u and GVC_u -based two-way authentication helps maintain forward secrecy. Since $\mathcal{A}$ cannot get the required secret information to generate the session key, it cannot obtain SK_u of previous sessions as well. Therefore, UAAS can also guarantee backward secrecy.

8) *Attainment of Untraceability*: Since only U and G are able to calculate DID_u based on R_u and N_u , an adversary cannot reveal the temporary identity of a UAV during an active session between U and G . The DID is also used to encode several messages. The challenge is not stored in G 's database and the final responses are generated using fuzzy extractor by both U and

TABLE III
SECURITY FEATURES COMPARISON

Feature	Alladi <i>et al.</i> [12]	Lei <i>et al.</i> [24]	Bansal <i>et al.</i> [11]	Pu <i>et al.</i> [13]	Wazid <i>et al.</i> [25]	UAAS
C5.1	✓	✓	✓	✓	✓	✓
C5.2	✓	✓	✓	✓	✓	✓
C5.3	✓	✓	✓	✓	✓	✓
C5.4	✓	✗	✓	✓	✗	✓
C5.5	✓	✗	✓	✓	✗	✓
C5.6	✓	✓	✗	✓	✓	✓
C7	✓	✗	✗	✓	✓	✓
C8	✓	✓	✗	✓	✗	✓
C10	✓	✗	✗	✓	✓	✓
C11	✓	✓	✗	✗	✓	✓
C12	✓	✓	✗	✓	✓	✓
Untraceability property	✗	✓	✗	✗	✓	✓
ESL attack resistance	✗	✗	✗	✗	✗	✓

G . Thus, without knowing DID_u , R_u , and N_u , it is infeasible for the adversary to trace both U and G , and consequently the untraceability property is preserved by UAAS.

9) *Resilience Against ESL Attacks*: At a given time, adversary $\mathcal{A}$ can derive SK_u only if DID_u , R_u , and N_u are exposed. Moreover, each session time is intelligently adapted such that the idle time of the session is minimized. Thus, $\mathcal{A}$ does not have sufficient time to expose SK_u of an established session. Therefore, UAAS can resist ESL attacks.

VI. COMPARATIVE STUDY

In this section, we provide a comparative analysis between UAAS and existing related mechanisms, such as the protocols proposed by Alladi *et al.* [12], Lei *et al.* [24], Bansal *et al.* [11], Pu *et al.* [13], and Wazid *et al.* [25].

A. Baseline Mechanisms

Authors in [12] propose a mutual authentication protocol based on PUFs for UAV-GS authentication, which is further extended for implementing UAV-UAV authentication. The work uses three nonce values and does not apply fuzzy extractor to regenerate responses. Considering the security risks in IoD networks, a security authentication protocol is designed in [24] for IoD networks. In that work, the Chinese residual theorem is applied for identity generation and verification, where 3 phase-design and 7 authentication processes are used to ensure forward and backward security, along with resistance to mainstream network attacks. Using PUFs, the work in [11] designs a mutual authentication scheme between a base station and UAVs. The K-Means clustering approach is utilized to form UAV clusters based on the locations of the UAVs. To provide secure communications between the GS and UAVs, a mutual authentication protocol, in which the PUF is engaged to generate the challenge-response pair to be used by UAVs and the GS as the condition of chaotic system, is proposed in [13]. Consequently, the message which is used to generate a secret key is shuffled randomly. Authors in [25] consider different security and privacy concerns in the deployment of IoD and accordingly design an authentication and secure key agreement mechanism between an accessed drone and a user with the help of a server device.

B. Security and Functionality Features Comparison

Table III presents a comparison of the security and functionality features, discussed in Section V-D. In Table III, '✓' denotes that the mechanism satisfies a criterion; whereas, '✗' indicates that the criterion is not satisfied by the mechanism. UAAS and baselines can protect against known attacks (C5.1)-(C5.3) and resist de-synchronization attacks (C5.6). The mechanisms also provide secure key agreement (C7), mutual authentication (C10), and attainment of forward secrecy (C12). By using PUFs, UAAS, [12], [11], and [13] offer security against node tampering (C5.4) and cloning attacks (C5.5). The mechanisms proposed in [24], [11], and [25] require synchronized clocks in the network, and thus fail the no clock synchronization feature (C8). In [13], since the ID of a UAV is not updated in a new session and is communicated in text between two parties, the user anonymity (C11) security feature is not satisfied. The untraceability property is satisfied by [24], [25], and UAAS. In our approach, the intelligent adaptation of the session key helps satisfy the ESL attack resistance that is not satisfied by baselines.

C. Computational Cost Comparison

We use T_h , T_f , T_e , and T_a to define the time required for computing hash operations, fuzzy extractor operations, encryption/decryption, and analog extractor operations, respectively. The costs associated with the cryptographic procedures corresponding to T_h , T_f , T_e , and T_a have been calculated in various existing works, such as [50], [51], and [24]. Since hash operations, fuzzy extractor operations, encryption/decryption, and analog extractor operations are generic cryptographic procedures, we have referenced the costs of these operations from the works mentioned in [50], [51], and [24]. Based on the experimental results used in [24], [50], [51], we have $T_h \approx 0.5$ ms, $T_f \approx 63.075$ ms, $T_e \approx 8.7$ ms, and $T_a \approx 2.045$ ms. As for the computational overhead, the PUF operation belongs to the sensor encryption computation, which has a low cost and is lightweight [24], [52]. Thus, we do not provide any cost estimation for the PUF computation.

Table IV presents the computational cost of UAAS and baselines. In UAAS, during the authentication process, G and each U need to perform $4T_h + T_f$ operations, which leads

TABLE IV
COMPUTATIONAL COST COMPARISON

Scheme	UAV	Ground station	User	Total cost
UAAS	$4T_h + T_f$ ≈ 65.075	$4T_h + T_f$ ≈ 65.075	-	$8T_h + 2T_f$ ≈ 130.15 ms
Alladi <i>et al.</i> [12]	$4T_h$ ≈ 2.0	$4T_h$ ≈ 2.0	-	$8T_h$ ≈ 4.0 ms
Lei <i>et al.</i> [24]	$2T_h$ ≈ 1.0	$4T_h + T_a$ ≈ 4.045	$2T_h$ ≈ 1.0	$8T_h + T_a$ ≈ 6.045 ms
Bansal <i>et al.</i> [11]	$2T_e$ ≈ 17.4	$2T_e$ ≈ 17.4	-	$4T_e$ ≈ 34.8 ms
Pu <i>et al.</i> [13]	$4T_e$ ≈ 34.8	$4T_e$ ≈ 34.8	-	$8T_e$ ≈ 69.6 ms
Wazid <i>et al.</i> [25]	$7T_h$ ≈ 3.5	$8T_h$ ≈ 4	$16T_h + T_f$ ≈ 71.075	$31T_h + T_f$ ≈ 78.575 ms

TABLE V
COMMUNICATION COST COMPARISON

Scheme	No. of messages	No. of bits
UAAS	3	1024
Alladi <i>et al.</i> [12]	3	1152
Lei <i>et al.</i> [24]	7	1536
Bansal <i>et al.</i> [11]	2	992
Pu <i>et al.</i> [13]	3	1632
Wazid <i>et al.</i> [25]	3	1696

to a total computational cost of $8T_h + 2T_f$ operations. This is approximately 130.15 ms, which is higher than baselines. This is because our mechanism uses fuzzy extractors, and a fuzzy extractor operation takes quite a bit more time than a hash, encryption/decryption, or analog extractor operation does. Although the mechanism proposed in Alladi *et al.* [12] has the lowest computational cost, it uses PUFs without applying a fuzzy extractor, and therefore suffers from PUF errors due to the impact of environmental variations on the digital circuit characteristics. Similarly, the mechanisms proposed in [13] and [11] use PUFs without considering fuzzy extractors. If we exclude the fuzzy extractor's operation time from the total cost, UAAS has a total computation cost of 4.0 ms, which is equal to that of the scheme in Alladi *et al.* [12]. Therefore, our proposed mechanism reduces the noise in PUFs with the help of fuzzy extractors, and as a result, its computational cost is higher than that of the baselines. The time costs of XOR and concatenation operations are significantly less than cryptographic operations. Thus, the overheads of XOR and concatenation operations are ignored in the proposed UAAS. In addition, since session time adaptation is performed after successful UAV-GS or UAV-UAV authentication, it is not included in the calculation of overheads of cryptographic operations in UAAS. However, we perform a time-bound analysis of session time adaptation in Section IV-E, which represents the computational cost required to dynamically set the session time.

D. Communication Cost Comparison

To calculate the communication cost, we follow the approach used in [53], which assumes that the hash output (using the secure hash algorithm SHA-1 [54]), the random number/nonce, and the identity consist of 160, 128, and 160 bits, respectively. Moreover, we assume that the PUF challenge and response are each 128 bits long, as specified in [53]. For the baselines, the timestamp, ECC point and the ciphertext block (considering Advanced Encryption Standard (AES-128) encryption) are assumed to be 32, 320, and 128 bits, respectively. In Table V,

TABLE VI
SIMULATION PARAMETERS

Parameter	Value
Area	$400m \times 400m \times 200m$
Wireless standard	IEEE 802.11ac
Channel bandwidth	40 MHz
Data and control mode	Constant rate wifi manager
Maximum physical data rate	41 Mbps
Path loss model	Log-normal path loss model (path loss exponent=3.0)
Propagation delay model	Constant speed propagation delay model
Simulation time	100 seconds
Number of repetitions of simulation for each number of station, for computing average performance	10
Mobility model	Random walk 2d mobility model ("Mode: Time", "Time: 2s", "Bounds: Rectangle (0, 200, 0, 200)", "Speed: UniformRandomVariable[Min=2.0, Max=4.0]", "Direction: UniformRandomVariable[Min=0.0, Max=6.283184]")
Value of δ	5.0 s
Maximum transmit power of UAVs	5 W
Noise power	-110 dBm

we present the communication cost comparison along with the number of messages communicated between the parties in the different mechanisms.

In UAAS, a total of 3 messages are exchanged between the UAV and GS during the UAV-GS authentication. The first $\langle DID_u, Q'_u, E_u, UVC_u \rangle$, second $\langle GVC_u \rangle$, and third messages $\langle SKVC_u, Res'_u \rangle$ require $160 + 128 + 128 + 160 = 576$ bits, 160 bits, and $160 + 128 = 288$ bits, respectively. Thus, the total communication cost of UAAS is 1024 bits. From Table V, it is noted that the scheme in [25] has the highest communication cost and the scheme in [24] requires the highest number of message exchanges. Our approach has the second lowest communication overhead, which is 32 bits higher than the scheme in [11]; however, it is not based on a two-way authentication and does not use hash functions.

VII. PRACTICAL IMPACT STUDY

We implement UAAS in NS-3.35 [38]. We consider 50 UAVs and one GS. The UAVs and GS are located at the center of the simulation area and communicate using the 5 GHz IEEE 802.11ac wireless standard. We measure the UDP throughput and compute the PLR, the end-to-end delay, and the probability of idle time in secure sessions. To compute these metrics, packets are transmitted from the GS to the UAVs and vice versa as well as between two UAVs. The throughput is calculated as the amount of data transmitted per unit of time. To compute the PLR, the number of packets that are not successfully received is divided by the number of packets that are successfully transmitted. The end-to-end delay represents the time it takes for one packet to be sent across the network from the source to the destination. Each simulation instance is repeated 10 times, and the aforementioned metrics are averaged over the 10 iterations. To calculate the probability of idle time in secure sessions, we compute the probability mass function (PMF) of the idle time in secure sessions. To this end, we run the simulation for 200 s and calculate the PMF of the idle time after each 10 s. Table VI details the simulation parameters. The parameters that are not explicitly mentioned in Table VI are assumed to take the default values specified by the NS-3.35 simulator.

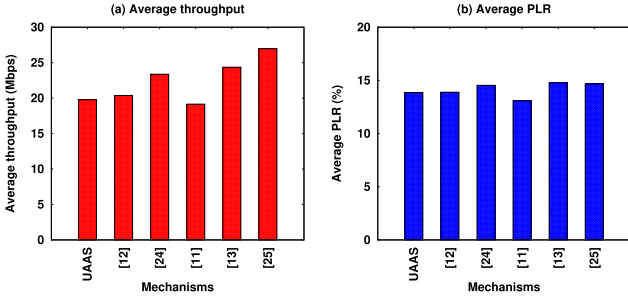


Fig. 6. (a) Throughput and (b) PLR

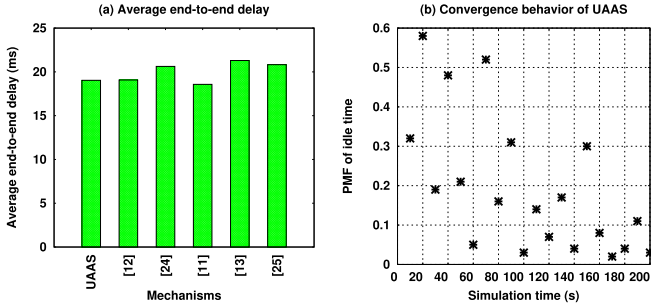


Fig. 7. (a) End-to-end delay and (b) Probability distribution of idle time in secure sessions

A. Throughput Analysis

Fig. 6(a) shows the throughput analysis, where the throughputs for UAAS are 20.38%, 23.37%, 24.36%, and 26.98% lower than those of Alladi et al. [12], Lei et al. [24], Pu et al. [13], and Wazid et al. [25], respectively. The throughput of UAAS is lower than that for baselines except [11]. This is because UAAS has less communication cost than [12], [24], [13], and [25], as shown in Table V.

B. PLR Analysis

Fig. 6(b) shows the PLR of UAAS and other mechanisms. Since higher communication overheads increase the probability of packet loss in a packet transmission period, from Fig. 6(b), it can be observed that there is a tendency to increase the PLR as the communication cost increases. UAAS has approximately 4.6%, 6.2%, and 5.5% lower PLRs than [24], [13], and [25], respectively; whereas, UAAS's PLR is comparable to the scheme in Alladi et al. [12].

C. End-to-End Delay Analysis

The results of the end-to-end delay are shown in Fig. 7(a). Due to the use of small sized messages, the required end-to-end delay is reduced in UAAS. As a result, the delay for UAAS is approximately 7.7%, 9%, and 8.6% lower than the schemes in [24], [13], and [25], respectively. Moreover, UAAS and the scheme in Alladi et al. [12] have similar end-to-end delays. Among baselines, since the smallest sized messages are used in [11], its end-to-end delay is approximately 2.4% lower than that of UAAS.

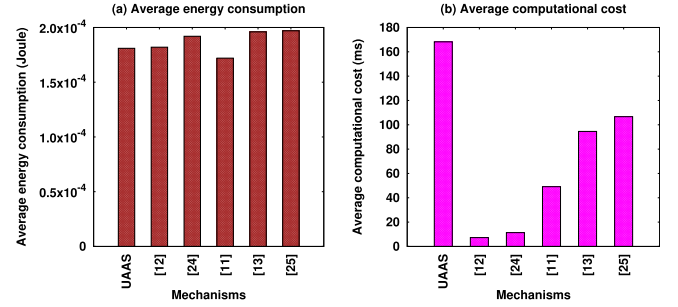


Fig. 8. (a) Energy consumption and (b) Computational cost

D. Probability of Idle Time in Secure Sessions

In an authenticated session, idle time denotes the amount of time when no communication takes place between the UAV and the GS. To compute the probability of idle time in secure sessions, we first compute the PMF of the idle time in authenticated sessions. We use the PMF of the session time to analyze the convergence of the session time adaptation process in UAAS. We run the simulation for 200 s and compute the PMF of the idle time of secure sessions each 10 s. Therefore, the PMF is calculated from the sum of the idle time in secure sessions after each 10 s run of the simulation. The result is shown in Fig. 7(b). It can be noted from this figure that the PMF of authenticated sessions decreases as the simulation time increases, which signifies that the learning-based session time adaptation converges because the session time is dynamically set so that the idle time of an active session is reduced. Due to the application of *exploration-exploitation* when selecting the session time, the statistical table $\mathcal{E}$ is enriched with more adaptive values as the execution time progresses. As a result, the PMF of the idle time decreases as the simulation progresses, as shown in Fig. 7(b), where it is noted that UAAS converges after approximately 80 s.

E. Energy Consumption and Computational Cost Analysis

As shown in Fig. 8(a), the average energy consumption of UAAS is approximately 0.55%, 5.73%, 7.65%, and 8.12% lower than that of the schemes in [12], [24], [13], and [25], respectively. This is because UAAS involves fewer bit transmissions than the aforementioned baselines. The results of the average computational cost are shown in Fig. 8(b). As discussed in Section VI-C, the computational cost of UAAS is higher than that of the baselines due to the use of the fuzzy extractor.

F. Discussion

Overall, UAAS provides a high level of security by maintaining the secrecy of the key elements and being secure against various well-known attacks, such as MM, masquerade, replay, node tampering, cloning, de-synchronization, and ESL attacks. In addition, by avoiding clock synchronization, UAAS offers several desirable features such as secure key agreement, mutual authentication, user anonymity, forward secrecy, and untraceability. Compared to the baselines, UAAS has a relatively low

communication cost, except for the scheme proposed in [11]. However, it uses fuzzy extractors, which increase its computational cost compared to baselines that do not use this technique. If we exclude the fuzzy extractor's operation time from the total cost, the computation cost of UAAS is comparable to that of the scheme in [12]. In the study of the practical impact of UAAS, it is noted that UAAS has a lower average throughput, PLR, and end-to-end delay than the baselines due to its lower communication overhead.

VIII. CONCLUSION

UAAS performs a two-way authentication-based secure session establishment for communication between a UAV and the GS. To ensure physical security, we jointly use PUFs and fuzzy extractors. In UAAS, the proposed UAV-GS authentication is further extended to create an authenticated session between two UAVs. A unique session key is generated for each session, which is used for secure communication between two parties. In order to minimize the unnecessary idle time in an active session, we intelligently set the session time of an authenticated session by using a TS-based online learning mechanism, which further helps enhance the security in UAV communications. The proposed mechanism does not store any secret keys and communicates important values, such as response and nonce values, in encrypted format. Through formal security proof, informal security analysis, and comparison with related protocols, it was shown that UAAS is secure and versatile. Based on the provided simulation results, it is also noted that UAAS significantly improves the network performance compared to baselines.

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REFERENCES

- [1] V. Chamola, V. Hassija, V. Gupta, and M. Guizani, "A comprehensive review of the COVID-19 pandemic and the role of IoT, drones, AI, blockchain, and 5G in managing its impact," *IEEE Access*, vol. 8, pp. 90225–90265, 2020.
- [2] K. P. Valavanis and G. J. Vachtsevanos, *Handbook of Unmanned Aerial Vehicles-5 Volume Set*. New York, NY, USA: Springer, 2014.
- [3] M. Singh, G. S. Aujla, and R. S. Bali, "A deep learning-based blockchain mechanism for secure internet of Drones environment," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 7, pp. 4404–4413, Jul. 2020.
- [4] B. Bera, S. Saha, A. K. Das, N. Kumar, P. Lorenz, and M. Alazab, "Blockchain-envisioned secure data delivery and collection scheme for 5G-based IoT-enabled Internet of Drones environment," *IEEE Trans. Veh. Technol.*, vol. 69, no. 8, pp. 9097–9111, Aug. 2020.
- [5] D. He, S. Chan, and M. Guizani, "Communication security of unmanned aerial vehicles," *IEEE Wireless Commun.*, vol. 24, no. 4, pp. 134–139, Aug. 2016.
- [6] R. Altawy and A. M. Youssef, "Security, privacy, and safety aspects of civilian drones: A survey," *ACM Trans. Cyber-Phys. Syst.*, vol. 1, no. 2, pp. 1–25, 2016.
- [7] C. Pu, "Jamming-resilient multipath routing protocol for flying ad hoc networks," *IEEE Access*, vol. 6, pp. 68472–68486, 2018.
- [8] T. Alladi, V. Venkatesh, V. Chamola, and N. Chaturvedi, "Drone-MAP: A novel authentication scheme for drone-assisted 5G networks," in *Proc. IEEE Conf. Comput. Commun. Workshops*, 2021, pp. 1–6.
- [9] K. Ranjitha et al., "Accelerating PUF-based UAV authentication protocols using programmable switch," in *Proc. 14th Int. Conf. Commun. Syst. Netw.*, 2022, pp. 309–313.
- [10] G. Bansal and B. Sikdar, "S-MAPS: Scalable mutual authentication protocol for dynamic UAV swarms," *IEEE Trans. Veh. Technol.*, vol. 70, no. 11, pp. 12088–12100, Nov. 2021.
- [11] G. Bansal and B. Sikdar, "Location aware clustering: Scalable authentication protocol for UAV swarms," *IEEE Netw. Lett.*, vol. 3, no. 4, pp. 177–180, Dec. 2021.
- [12] T. Alladi, Naren, G. Bansal, V. Chamola, and M. Guizani, "SecAuthUAV: A novel authentication scheme for UAV-ground station and UAV-UAV communication," *IEEE Trans. Veh. Technol.*, vol. 69, no. 12, pp. 15068–15077, Dec. 2020.
- [13] C. Pu and Y. Li, "Lightweight authentication protocol for unmanned aerial vehicles using physical unclonable function and chaotic system," in *Proc. IEEE Int. Symp. Local Metrop. Area Netw.*, 2020, pp. 1–6.
- [14] C. Pu, A. Wall, K.-K. R. Choo, I. Ahmed, and S. Lim, "A lightweight and privacy-preserving mutual authentication and key agreement protocol for Internet of Drones environment," *IEEE Internet Things J.*, vol. 9, no. 12, pp. 9918–9933, Jun. 2022.
- [15] M. Zhang, C. Xu, S. Li, and C. Jiang, "On the security of an ECC-based authentication scheme for Internet of Drones," *IEEE Syst. J.*, vol. 16, no. 4, pp. 6425–6428, Dec. 2022.
- [16] S. Hussain, S. A. Chaudhry, O. A. Alomari, M. H. Alsharif, M. K. Khan, and N. Kumar, "Amassing the security: An ECC-based authentication scheme for Internet of Drones," *IEEE Syst. J.*, vol. 15, no. 3, pp. 4431–4438, Sep. 2021.
- [17] C. Feng, B. Liu, Z. Guo, K. Yu, Z. Qin, and K.-K. R. Choo, "Blockchain-based cross-domain authentication for intelligent 5G-enabled Internet of Drones," *IEEE Internet Things J.*, vol. 9, no. 8, pp. 6224–6238, Apr. 2022.
- [18] S. Yu, A. K. Das, Y. Park, and P. Lorenz, "SLAP-IoD: Secure and lightweight authentication protocol using physical unclonable functions for Internet of Drones in smart city environments," *IEEE Trans. Veh. Technol.*, vol. 71, no. 10, pp. 10374–10388, Oct. 2022.
- [19] K. Lounis, S. H. Ding, and M. Zulkernine, "D2D-MAP: A drone to drone authentication protocol using physical unclonable functions," *IEEE Trans. Veh. Technol.*, vol. 72, no. 4, pp. 5079–5093, Apr. 2023.
- [20] M. Tanveer et al., "RAMP-IoD: A robust authenticated key management protocol for the Internet of Drones," *IEEE Internet Things J.*, vol. 9, no. 2, pp. 1339–1353, Jan. 2021.
- [21] Y. Tan, J. Wang, J. Liu, and N. Kato, "Blockchain-assisted distributed and lightweight authentication service for industrial unmanned aerial vehicles," *IEEE Internet Things J.*, vol. 9, no. 18, pp. 16928–16940, Sep. 2022.
- [22] G. Raja, S. G. Senthivel, S. Balaganesh, B. Rajakumar, V. Ravichandran, and M. Guizani, "MLB-IoD: Multi layered blockchain assisted 6G Internet of Drones ecosystem," *IEEE Trans. Veh. Technol.*, vol. 72, no. 2, pp. 2511–2520, Feb. 2023.
- [23] M. W. Akram et al., "A secure and lightweight drones-access protocol for smart city surveillance," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 10, pp. 19634–19643, Oct. 2022.
- [24] Y. Lei, L. Zeng, Y.-X. Li, M.-X. Wang, and H. Qin, "A lightweight authentication protocol for UAV networks based on security and computational resource optimization," *IEEE Access*, vol. 9, pp. 53769–53785, 2021.
- [25] M. Wazid, A. K. Das, N. Kumar, A. V. Vasilakos, and J. J. Rodrigues, "Design and analysis of secure lightweight remote user authentication and key agreement scheme in Internet of Drones deployment," *IEEE Internet Things J.*, vol. 6, no. 2, pp. 3572–3584, Apr. 2019.
- [26] S. U. Jan, F. Qayum, and H. U. Khan, "Design and analysis of lightweight authentication protocol for securing IoD," *IEEE Access*, vol. 9, pp. 69287–69306, 2021.
- [27] Z. Ali, S. A. Chaudhry, M. S. Ramzan, and F. Al-Turjman, "Securing smart city surveillance: A lightweight authentication mechanism for unmanned vehicles," *IEEE Access*, vol. 8, pp. 43711–43724, 2020.
- [28] C. Jiang, Y. Fang, P. Zhao, and J. Panneerselvam, "Intelligent UAV identity authentication and safety supervision based on behavior modeling and prediction," *IEEE Trans. Ind. Inform.*, vol. 16, no. 10, pp. 6652–6662, Oct. 2020.
- [29] M. A. Khan et al., "A provable and privacy-preserving authentication scheme for UAV-enabled intelligent transportation systems," *IEEE Trans. Ind. Inform.*, vol. 18, no. 5, pp. 3416–3425, May 2022.

- [30] S. J. Maeng, Y. Yapici, I. Guvenc, A. Bhuyan, and H. Dai, "Precoder design for physical-layer security and authentication in massive MIMO UAV communications," *IEEE Trans. Veh. Technol.*, vol. 71, no. 3, pp. 2949–2964, Mar. 2022.
- [31] T. Li et al., "Energy-efficient and secure communication towards UAVs networks," *IEEE Internet Things J.*, vol. 9, no. 12, pp. 10061–10076, Jun. 2022.
- [32] P. Gope, A. K. Das, N. Kumar, and Y. Cheng, "Lightweight and physically secure anonymous mutual authentication protocol for real-time data access in industrial wireless sensor networks," *IEEE Trans. Ind. Inform.*, vol. 15, no. 9, pp. 4957–4968, Sep. 2019.
- [33] W. R. Thompson, "On the likelihood that one unknown probability exceeds another in view of the evidence of two samples," *Oxford J.*, vol. 25, no. 3/4, pp. 285–294, Dec. 1933.
- [34] J.-P. Linnartz and P. Tuyls, "New shielding functions to enhance privacy and prevent misuse of biometric templates," in *Proc. Int. Conf. Audio Video-Based Biometric Person Authentication*, Springer, 2003, pp. 393–402.
- [35] W. Mao and C. Boyd, "Towards formal analysis of security protocols," in *Proc. Comput. Secur. Found. Workshop VI*, 1993, pp. 147–158.
- [36] M. Abdalla, P.-A. Fouque, and D. Pointcheval, "Password-based authenticated key exchange in the three-party setting," in *Proc. 8th Int. Workshop Theory Pract. Public Key Cryptogr.*, Les Diablerets, Switzerland, Springer, 2005, pp. 65–84.
- [37] "The scyther tool," 2016, Accessed: May 10, 2022. [Online]. Available: <https://people.cispa.io/cas.cremers/scyther/>
- [38] "Ns-3.35 - nsnam." [Online]. Available: <https://www.nsnam.org/releases/ns-3-35>
- [39] S. Devadas, E. Suh, S. Paral, R. Sowell, T. Ziola, and V. Khandelwal, "Design and implementation of PUF-based "unclonable" RFID ICs for anti-counterfeiting and security applications," in *Proc. IEEE Int. Conf. RFID*, 2008, pp. 58–64.
- [40] P. Gope, J. Lee, and T. Q. Quek, "Lightweight and practical anonymous authentication protocol for RFID systems using physically unclonable functions," *IEEE Trans. Inf. Forensics Secur.*, vol. 13, no. 11, pp. 2831–2843, Nov. 2018.
- [41] J. H. Cheon, J. Jeong, D. Kim, and J. Lee, "A reusable fuzzy extractor with practical storage size: Modifying Canetti et al.'s construction," in *Proc. Australas. Conf. Inf. Secur. Privacy*, Springer, 2018, pp. 28–44.
- [42] C. Watkins, "Learning from delayed rewards," PhD thesis, University of Cambridge, Cambridge, U.K., May 1989.
- [43] P. Auer, N. Cesa-Bianchi, and P. Fischer, "Finite-time analysis of the multiarmed bandit problem," *J. Mach. Learn.*, vol. 47, no. 2, pp. 235–256, May 2002.
- [44] D. Wang, D. He, P. Wang, and C.-H. Chu, "Anonymous two-factor authentication in distributed systems: Certain goals are beyond attainment," *IEEE Trans. Dependable Secure Comput.*, vol. 12, no. 4, pp. 428–442, Jul./Aug. 2014.
- [45] D. Wang, H. Cheng, D. He, and P. Wang, "On the challenges in designing identity-based privacy-preserving authentication schemes for mobile devices," *IEEE Syst. J.*, vol. 12, no. 1, pp. 916–925, Mar. 2018.
- [46] S. Roy, A. K. Das, S. Chatterjee, N. Kumar, S. Chattopadhyay, and J. J. Rodrigues, "Provably secure fine-grained data access control over multiple cloud servers in mobile cloud computing based healthcare applications," *IEEE Trans. Ind. Inform.*, vol. 15, no. 1, pp. 457–468, Jan. 2019.
- [47] C. J. Cremers, P. Lafourcade, and P. Nadeau, "Comparing state spaces in automatic security protocol analysis," in *Formal to Practical Security*. Berlin, Germany: Springer, 2009, pp. 70–94.
- [48] D. Wang and P. Wang, "Two birds with one stone: Two-factor authentication with security beyond conventional bound," *IEEE Trans. Dependable Secure Comput.*, vol. 15, no. 4, pp. 708–722, Jul./Aug. 2018.
- [49] D. Wang, W. Li, and P. Wang, "Measuring two-factor authentication schemes for real-time data access in industrial wireless sensor networks," *IEEE Trans. Ind. Inform.*, vol. 14, no. 9, pp. 4081–4092, Sep. 2018.
- [50] D. He, N. Kumar, M. K. Khan, and J.-H. Lee, "Anonymous two-factor authentication for consumer roaming service in global mobility networks," *IEEE Trans. Consum. Electron.*, vol. 59, no. 4, pp. 811–817, Nov. 2013.
- [51] Q. Jiang, J. Ma, G. Li, and L. Yang, "An efficient ticket based authentication protocol with unlinkability for wireless access networks," *Wireless Pers. Commun.*, vol. 77, no. 2, pp. 1489–1506, 2014.
- [52] A. Babaei and G. Schiele, "Physical unclonable functions in the Internet of Things: State of the art and open challenges," *Sensors*, vol. 19, no. 14, 2019, Art. no. 3208.
- [53] S. Banerjee, V. Odelu, A. K. Das, S. Chattopadhyay, J. J. Rodrigues, and Y. Park, "Physically secure lightweight anonymous user authentication protocol for internet of things using physically unclonable functions," *IEEE Access*, vol. 7, pp. 85627–85644, 2019.
- [54] S. H. Standard, "FIPS Pub 180–1," *Nat. Inst. Standards Technol.*, vol. 17, no. 180, 1995, Art. no. 15.



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